1. Mobile Computing

1. Mobile computing enables communication, computation, and sensing for physically mobile users and devices.
2. Mobile computing mainly relies on wired connectivity as a key characteristic.
3. In ubiquitous computing, the computer disappears into the background.
4. Pervasive computing involves embedding computing into everyday objects with continuous data exchange.
5. Mobile computing systems must operate under changing network conditions such as WiFi, LTE, and 5G.
6. Mobile devices typically have unlimited CPU, memory, and energy resources.
7. Which of the following is NOT a characteristic of mobile computing?

   A. Mobility
   B. Context awareness
   C. Distributed architecture
   D. Fixed-location-only computing

8. Which of the following is a major challenge in mobile computing?

   A. Unlimited battery capacity
   B. Stable and fixed network conditions
   C. Security and privacy issues
   D. Infinite computational resources

9. What was a key turning point in the smartphone revolution (2007–2012)?

   A. Coda file system
   B. iPhone and Android ecosystem
   C. Cloudlets
   D. Model compression

10. Which of the following is NOT listed as a smartphone sensor in mobile sensing?

    A. Accelerometer
    B. Gyroscope
    C. GPS
    D. LiDAR

11. Which of the following is NOT a research direction for deep learning on mobile?

    A. Model compression
    B. Quantization
    C. Hardware acceleration
    D. Desktop-only processing

12. What is the main idea of **cloud offloading** in mobile computing?

    A. Increasing battery size
    B. Moving computation to remote servers
    C. Eliminating network usage
    D. Running everything locally only

13. What is the **core idea of ubiquitous computing**?

    A. High-performance centralized computing
    B. Computers become invisible and integrated into the environment
    C. Devices are always manually controlled
    D. All computation happens in the cloud

14. Which of the following best describes **mobility in mobile computing**?

    A. Devices remain stationary
    B. Devices frequently change physical location and network attachment
    C. Only cloud servers move
    D. Only data moves, not devices

15. Which of the following are **resource constraints** in mobile computing?

    A. Limited battery
    B. Limited CPU/memory
    C. Unlimited storage
    D. Limited bandwidth

16. Which of the following are **characteristics of wireless communication** in mobile computing?

    A. Variable bandwidth
    B. Intermittent connectivity
    C. Fixed latency
    D. Variable latency

17. Name **two resource constraints** in mobile computing.

18. What are **two key characteristics** of wireless communication in mobile computing?

19. What is one **core feature of on-device AI (2020s)** in mobile computing?

20. What is one **example of adaptation** required in mobile systems?

2. On-Device AI

1. On-device AI performs inference on the same device where data is generated.
2. Cloud-based AI always guarantees low latency.
3. Edge computing eliminates the need for any network connection.
4. On-device AI can reduce bandwidth usage.
5. Increasing compute always improves sustained performance on devices.
6. Thermal throttling can increase latency variability.
7. DRAM access is more energy-efficient than computation.
8. Peak TOPS is always the best indicator of real performance.
9. Which are benefits of on-device AI?
   A. Privacy
   B. Reduced latency
   C. Zero networking cost
   D. Infinite compute

10. Which are characteristics of cloud-based AI?
    A. High latency
    B. Massive compute resources
    C. Zero bandwidth cost
    D. Single point of failure

11. Which are system constraints in on-device AI?
    A. Limited compute
    B. Limited energy
    C. Unlimited cooling
    D. Limited memory

12. Which components are part of a smartphone SoC?
    A. CPU B. GPU C. NPU D. PSU

13. Which statements about CPU are correct?
    A. Handles general-purpose control
    B. Good for irregular operations
    C. Most energy-efficient for tensor ops
    D. Used for orchestration

14. Which statements about GPU are correct?
    A. High parallel throughput B. Low power consumption
    C. Used for graphics D. Can run AI workloads

15. Which statements about NPU are correct?
    A. Optimized for tensor operations
    B. High TOPS per watt
    C. Handles all control logic
    D. Limited operator flexibility

16. Which contribute to peak memory usage?
    A. Model weights B. Activations C. Runtime buffers D. Display memory

17. Which are reasons memory is a bottleneck?
    A. DRAM access is slow
    B. DRAM access is energy expensive
    C. SRAM is unlimited
    D. Memory is shared across components

18. Which techniques reduce memory cost?
    A. Tiling
    B. Keeping data in SRAM
    C. Increasing DRAM access
    D. Reducing data movement

19. Which are steps in deployment pipeline?
    A. Training B. Quantization C. Export to IR D. Compilation

20. Which are examples of IR formats?
    A. ONNX B. TFLite C. Core ML D. CUDA

21. Which are goals of quantization?
    A. Reduce model size
    B. Reduce energy consumption
    C. Increase precision
    D. Enable NPU execution

22. Which are runtime responsibilities?
    A. Load model B. Manage execution C. Allocate buffers D. Train model

23. SoC stands for ______ on Chip.

24. ______ latency refers to worst-case latency (e.g., 99th percentile).

25. The ______ cache in LLMs grows with context length.

26. ______ memory is fast, small, and energy-efficient on-chip memory.

27. ______ memory is large but slower and energy-expensive.

28. Quantization often converts FP16 or FP32 into ______.

29. The primary goal in energy optimization is minimizing energy per ______.

30. On-device AI is a co-design of Model + ______ + Hardware.

3. Model Compression

1. Model compression is only about reducing model size.
2. Edge devices have limited compute, memory, and energy resources.
3. Quantization always improves accuracy.
4. PTQ does not require retraining.
5. QAT uses fake quantization during forward pass.
6. Structured pruning is more hardware-friendly than unstructured pruning.

7. Which is NOT a main compression method?
   A. Quantization B. Pruning C. Knowledge Distillation D. Data Augmentation

8. What is the main purpose of quantization?
   A. Increase precision B. Reduce bit-width C. Increase model depth D. Add redundancy

9. Which format is commonly used for efficient inference?
   A. FP64 B. FP32 C. INT8 D. FP128

10. What does W8A8 mean?
    A. Weight FP8, Activation FP8
    B. Weight INT8, Activation INT8
    C. Weight FP16, Activation FP16
    D. Weight INT4, Activation INT8

11. Which is a drawback of PTQ?
    A. Requires retraining B. Slow
    C. Accuracy drop at low bit-width D. Large memory

12. What is removed in pruning?
    A. Data samples B. Loss functions
    C. Parameters or structures D. Input features

13. In distillation, the teacher model is usually:
    A. Smaller and faster B. Larger and more accurate
    C. Randomly initialized D. Sparse

14. What replaces attention in Mamba?
    A. CNN B. RNN C. State Space Model D. GAN

15. Which are benefits of quantization?
    A. Smaller model size
    B. Lower memory bandwidth
    C. Higher energy consumption
    D. Faster inference

16. Which factors are optimization goals in model compression?
    A. Accuracy B. Latency C. Memory footprint D. Energy

17. What can be compressed in a model?
    A. Weights B. Activations
    C. Bit precision D. Number of operations

18. Which are characteristics of weight-only quantization?
    A. Simpler B. Common in LLM inference
    C. Always best accuracy D. Quantizes activations

19. Which are disadvantages of unstructured pruning?
    A. Irregular sparsity
    B. Hard to accelerate
    C. Requires special kernels
    D. Too coarse-grained

20. Which are examples of structured pruning?
    A. Removing channels
    B. Removing filters
    C. Removing layers
    D. Removing individual weights

21. Which are properties of QAT?
    A. Simulates low precision during training
    B. Uses STE
    C. Always worse than PTQ
    D. Adapts model to quantization

22. Which are signals used in knowledge distillation?
    A. Ground-truth labels B. Teacher predictions
    C. Random noise D. Hard-coded rules

23. Which are advantages of knowledge distillation?
    A. Improves small model performance B. Helps recover accuracy
    C. Reduces training time always D. Works well with pruning/quantization

24. Which are objectives in NAS?
    A. Accuracy B. FLOPs C. Latency D. Parameter count

25. Quantization reduces ______ to make models smaller and faster.

26. In the equation q=round(r/S)+Z, **Z** is called the ______.

    q=round(r/S)+Zq = round(r/S) + Z

27. PTQ uses ______ data to estimate value ranges.

28. In pruning, removing weights with small values is called ______-based pruning.

29. In knowledge distillation, the small model is called the ______ model.

30. Mamba achieves ______-time complexity with respect to sequence length.

4. Adaptation & Personalization

1. Adaptation adjusts a model to a new domain or condition.
2. Personalization is a subset of adaptation.
3. Generalization requires additional fine-tuning to work on unseen data.
4. Domain shift means the data distribution changes between training and deployment.
5. A single model can perfectly handle all users, environments, and devices.
6. Test-Time Adaptation (TTA) uses incoming test data during deployment.
7. What is the **main challenge of domain shift**?

   A. Model becomes faster
   B. Accuracy drops when distribution changes
   C. Training becomes easier
   D. Data size decreases

8. Which is the **most common type of domain shift**?

   A. Label shift
   B. Concept shift
   C. Covariate shift
   D. Temporal shift

9. In domain adaptation, what is the **goal**?

   A. Increase model size
   B. Transfer knowledge from source to target domain
   C. Reduce dataset size
   D. Remove noise

10. Which method requires **labeled target data**?

    A. Unsupervised domain adaptation
    B. Supervised domain adaptation
    C. TTA
    D. ICL

11. What does **DANN** aim to achieve?

    A. Increase model parameters
    B. Align feature distributions across domains
    C. Remove labels
    D. Reduce input size

12. Which of the following is **NOT a PEFT method**?

    A. Adapter tuning
    B. LoRA
    C. Prefix tuning
    D. Full fine-tuning

13. What is updated in LoRA?

A. Entire weight matrix
B. Only input tokens
C. Low-rank matrices
D. Labels

14. What is the key limitation of ICL?

A. Requires gradient updates
B. Needs labeled data
C. Limited context window
D. Cannot use prompts

15. Which are reasons adaptation is necessary?

A. Deployment differs from training
B. Users have different preferences
C. Environments change over time
D. Models are always perfect

16. Which factors contribute to domain shift?

A. Different users
B. Different sensors/devices
C. Different environments
D. Same dataset reused

17. Which are types of domain shift?

A. Covariate shift
B. Label shift
C. Concept shift
D. Optimization shift

18. Which are characteristics of PEFT?

A. Update only a small subset of parameters
B. Reduce computational cost
C. Require full retraining
D. Suitable for large models

19. Which methods belong to PEFT?

A. Adapter
B. LoRA
C. Prompt tuning
D. SGD

20. Which statements about **ICL** are correct?

    A. No parameter updates
    B. Uses examples in prompts
    C. Requires retraining
    D. Can include user profile

21. Which are advantages of **fine-tuning**?

    A. Strong performance
    B. Simple to implement
    C. No computation cost
    D. Flexible across tasks

22. Which are **limitations of full fine-tuning**?

    A. Requires labeled target data
    B. Computationally expensive
    C. Always overfits
    D. Hard on edge devices

23. Which describe **Test-Time Adaptation (TTA)**?

    A. Happens during deployment
    B. Uses test data for adaptation
    C. Requires labels
    D. No retraining phase

24. Which are **common TTA techniques**?

    A. Entropy minimization
    B. Normalization statistics update
    C. Pseudo-labeling
    D. Dropout removal only

25. Domain shift is also called __________ shift.

26. In covariate shift, __________ changes but __________ stays the same.

27. In LoRA, the pretrained weights are __________ and only __________ updates are learned.

28. In ICL, instead of changing weights, we change the __________.

29. TTA adapts the model during __________ using incoming test data.

30. One major challenge of TTA is high __________ during inference.

5. Cloud-Edge Collaborative AI

1. Cloud-only systems eliminate all privacy concerns.
2. Edge-only systems can always handle large AI models efficiently.
3. Federated Learning sends model updates instead of raw data.
4. Centralized training requires collecting user data in one place.
5. Split inference always reduces communication cost.
6. Device and edge boundaries can sometimes be unclear.
7. Communication is often a bottleneck in FL.
8. SLMs generally provide deeper reasoning than LLMs.
9. Why is cloud-edge collaboration needed?

   A. Devices have unlimited resources
   B. Cloud has strong compute power
   C. Devices provide privacy and low latency
   D. Cloud eliminates all communication cost

10. Which are limitations of cloud-only AI?

    A. High latency
    B. Privacy concerns
    C. Unlimited compute
    D. Dependency on connectivity

11. Which are limitations of edge-only AI?

    A. Limited model size
    B. Limited energy
    C. Unlimited memory
    D. Thermal constraints

12. Which are system trade-offs in collaborative AI?

    A. Latency
    B. Energy
    C. Bandwidth
    D. UI design

13. In Federated Learning, which are true?

    A. Data stays on device
    B. Server aggregates updates
    C. Raw data is uploaded
    D. Clients perform local training

14. Which are steps in one FL round?

    A. Server selects clients
    B. Clients train locally
    C. Clients send updates
    D. Server discards updates

15. Which are FL challenges?

A. Statistical heterogeneity
B. System heterogeneity
C. Communication cost
D. Infinite bandwidth

16. Which relate to statistical heterogeneity?

A. Different user data distributions
B. Client drift
C. Same data across all devices
D. Conflicting objectives

17. Which relate to system heterogeneity?

A. Different battery levels
B. Different network quality
C. Same hardware across devices
D. Device availability differences

18. In split inference, which are true?

A. Model is partitioned
B. Entire model runs on device
C. Intermediate features are transmitted
D. Cloud always runs first layers

19. What factors affect split point selection?

A. Device speed
B. Network bandwidth
C. Activation size
D. User wallpaper

20. In SLM-LLM collaboration, which are valid patterns?

A. Draft (SLM) → Verify (LLM)
B. Plan (LLM) → Execute (SLM)
C. Perception (SLM) → Reasoning (LLM)
D. LLM always replaces SLM

21. Federated Learning keeps data ______ and trains a global model collaboratively.

22. In FL, the server performs ______ of client updates.

23. Split inference sends ______ features from device to server.

24. Speculative decoding uses a small model to generate ______ tokens.

25. In Sketch-Then-Fill, the LLM generates a high-level ______.

26. Name one advantage of on-device AI.

27. What is one drawback of centralized training?

28. What is the key idea of split inference?

29. Name one challenge of FL related to communication.

30. Why combine SLM and LLM instead of using only one?

6. Mobile LLMs VLMs VLAs Agent

1. LLMs can directly execute physical actions in the real world.
2. VLMs combine vision and language inputs.
3. VLA models generate robot actions as outputs.
4. Agents operate in a single-step manner.
5. On-device LLMs are suitable for offline use.
6. On-device latency is always lower than cloud latency.
7. Autoregressive decoding contributes to latency.
8. Agents interact with environments in a closed-loop manner.
9. What is the primary output of an LLM?

   A. Image
   B. Action
   C. Text
   D. Sensor signal

10. Which model introduces visual perception?

    A. LLM
    B. VLM
    C. VLA
    D. Agent

11. Which concept represents a system rather than a single model?

    A. LLM
    B. VLM
    C. VLA
    D. Agent

12. What is the main role of a vision encoder?

    A. Generate text
    B. Convert images into features
    C. Plan actions
    D. Execute commands

13. What is a key constraint of on-device AI?

    A. Unlimited compute
    B. High latency tolerance
    C. Limited memory and compute
    D. Infinite bandwidth

14. Which model maps vision + language to actions?

    A. LLM
    B. VLM
    C. VLA
    D. Agent

15. Which stage comes right after planning in an agent pipeline?

A. Tokenization
B. Action selection
C. Training
D. Compression

16. Which of the following are characteristics of VLMs?

A. Use vision encoder
B. Output only actions
C. Combine image and text
D. Perform multimodal reasoning

17. Which are challenges of on-device LLMs?

A. Autoregressive latency
B. Unlimited GPU memory
C. KV cache overhead
D. Energy constraints

18. Which components are part of a VLM pipeline?

A. Vision encoder
B. Projector
C. Text tokenizer
D. Action de-tokenizer

19. Which statements about Agents are correct?

A. Operate in loops
B. Perform planning
C. Always single-step
D. Use environment feedback

20. Which factors affect user experience of on-device LLMs?

A. Time-to-first-token
B. Throughput
C. Memory usage
D. Screen brightness

21. Which are differences between VLM and VLA?

A. VLA outputs actions
B. VLM outputs only actions
C. VLM focuses on perception
D. VLA connects to physical control

22. Which are parts of a mobile agent pipeline?

A. Perception
B. Planning
C. Execution
D. Reflection

23. Which describe Physical AI?

A. Only text processing
B. Perception + reasoning + action loop
C. Interaction with real-world environment
D. No sensors required

24. Which are difficulties of VLM compared to LLM?

A. Larger token budget
B. Vision encoding cost
C. Multimodal fusion overhead
D. No latency issues

25. LLM → VLM → VLA → Agent represents a progression from understanding to ______.

26. In VLM, images are first processed by a ______ encoder.

27. The ______ maps visual features into the language embedding space.

28. In LLM inference, ______ cache stores past key-value pairs.

29. Agents operate in a ______-loop system rather than one-shot prediction.

30. The action ______ converts discrete tokens into continuous robot commands.